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MILITARY SPECIFICATION

COUNTER, ROTATING, MECHANICAL, DIRECT DRIVE: 8587543 (3200 MIL RESET)

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the design, performance and inspection requirements for a direct drive, mechanical, high-speed reset counter used in fire control equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-L-6085	-	Lubricating Oil; Instrument, Aircraft, Low Volatility
MIL-L-7870	-	Lubricating Oil; General Purpose, Low Temperature
MIL-L-11734	-	Lubricating Oil; Synthetic (For Mechanical Time Fuzes)
MIL-F-13926	-	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1290

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MIL-P-14232 - Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-810 - Environmental Test Methods
MS33558 - Numerals and Letters, Aircraft Instrument Dial, Standard Form of

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

U.S. Army Armament, Research, Development and Engineering Center (ARDEC)

D8587543 - Counter, Rotating, Mechanical, Direct Drive (3200 MIL Reset)

PACKAGING DATA SHEET

8587543 - Packaging of Counter: 8587543

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Construction.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.2 and 6.2).

3.1.2 General description. The counter shall contain a mechanism that registers consecutive readings from 0000 to 9999 and repeats. It shall permit manual resetting to a 3200 reading from any random reading.

3.1.3 Fabrication. The counter shall be fabricated within the configuration shown on Drawing D8587543 and shall conform to the dimensional requirements shown thereon. A frame shall rigidly support the mechanism. The mechanism may be partially enclosed. Openings shall be provided for viewing and illuminating the digits and graduations but no transparent coverings are permitted.

3.1.4 Digits and graduations and index line. Digits shall be .187 ± .005 in. high, .099 ± .005 in. wide and .025 ± .005 line stroke. Digit (numeral) styling shall conform to MS33558 and shall all be of uniform quality. Imprinting of digits and graduations shall be performed by one process. The units wheel, in addition to the digits, shall contain 40 equally spaced graduations .020 ± .005 in. line stroke. Length and location of graduations shall be as specified on Drawing D8587543, however, the minimum visible length of the shortest graduation and index line shall be 3/32 inch.

3.1.5 Mask. The mask shall consist of individual apertures which shall permit illumination and viewing of the digits and graduations and shall be properly shaped and located so the markings are centralized within their respective openings. The mask shall be in close proximity to the numbered wheels to preclude ambiguous readings when viewed at a distance of 10 inches ± 2 inches and at an angle of 45° ± 5° to the reading line. An index line .020 ± .005 inches line stroke shall be located on the surface of the mask adjacent to the extreme right aperture and shall coincide with the zero graduation line when the zero digit of the unit wheel is on the reading line.

3.1.6 Reset mechanism. Resetting shall be accomplished by activating the reset shaft with a push in and turn motion on a counterclockwise direction. The push in distance shall not exceed .125 inch. Rotation shall not exceed one revolution. The reset shaft shall be positively locked upon completion of the reset cycle to prevent accidental activation.

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3.1.7 Materials.

3.1.7.1 Quality. All materials shall be of uniform quality and free of defects which may impair the functioning or service life of the counter.

3.1.7.2 Corrosion resistance. Metals and protective finishes shall be selected for maximum corrosion resistance. Dissimilar metals which have a tendency to produce galvanic corrosion shall not be used in intimate contact unless suitably proctected against such corrosion.

3.1.7.3 Fungus resistance. Materials shall be inherently fungistatic or treated with a suitable permanent fungicidal agent.

3.1.8 Final finishes.

3.1.8.1 Digits, index and graduations. Digits, index and graduations shall be white on a black background.

3.1.8.2 Exterior. The exterior finish of the frame, mask and case (if used) shall be black.

3.1.8.3 Colors. White and black finishes that may be amalgamated in the material, painted, inpregnated, hot stamped, or applied chemically, shall be in accordance with MIL-STD-171.

3.1.9 Lubrication. The counter components, where required, shall be lubricated very sparingly with oil that conforms to MIL-L-6085 or MIL-L-7870 or MIL-L-11734 or grease that conforms to MIL-G-23827. The counter shall meet all the requirements of 3.4 without requiring relubrication.

3.1.10 Identification markings. Part numbering shall be in accordance with MIL-STD-130. Serial numbers for use in identification marking will be assigned to the contractor by the procuring agency and shall be applied in the vicinity of the part number. Identification markings shall be visible on the exterior of the finished counter.

3.2 Environmental.

3.2.1 Storage temperatures. The counter shall show no evidence of physical failure and shall meet all the requirements of this specification at the standard ambient temperature after having been exposed to and thermally stabilized at ambient temperature of -80° and +160°F.

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3.2.2 Operating temperatures. The counter shall show no evidence of physical failure and shall meet the requirements of 3.3.4, 3.3.5, and 3.3.7 while thermally stabilized at ambient temperatures of -45° and +150°F.

3.2.3 Vibration. The counter shall show no evidence of physical failure (loose or broken parts) after being vibrated at a frequency of 30cps at an amplitude of 1/16 inch (1/8 inch total excursion) for five (5) minutes along each of the three (3) mutually perpendicular axes.

3.2.4 Shock. The counter shall show no evidence of physical failure (loose or broken parts) after being subjected to eighteen (18) impacts (total) of 100G's acceleration with each shock pulse having a time duration of 6 ± 1 milliseconds. Three (3) shocks to be applied in each direction along the three (3) mutually perpendicular axes.

3.3 Performance.

3.3.1 Operation. The counter shall add ten (10) counts for each clockwise revolution of the drive shaft and subtract ten (10) counts of opposite rotation.

3.3.2 Drive shaft. There shall be no binding or slipping of the drive shaft. Setting shall be easily attainable at any graduation setting.

3.3.3 Digit alignment. When any digit on the unit wheel is aligned on the reading line, the other three digits, tens, hundreds and thousands shall each be aligned on the reading line within 1/32 inch. This requirement shall be met at any random reading except when wheel transfer occurs between a 9 and zero (0). At the completion of wheel transfer between 9 and zero (0) or zero (0) and 9, the maximum misalignment of any one digit with respect to the reading line shall not exceed 0.50 inch. This requirement shall be met with backlash removed.

3.3.4 Speed. The counter shall show no evidence of miscounting when operated at 2500 RPM for 2 minute periods in both the clockwise and counterclockwise directions, starting each cycle from a dead stop position.

3.3.5 Reset operation. The counter shall reset to "3200" plus or minus one graduation from any random reading.

3.3.6 Markings. All markings shall be clearly defined at a distance of 10 inches ± 2 inches when viewed by a person with normal vision (correction permitted) subsequent to meeting the requirements of 3.4.

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3.3.7 Operability.

3.3.7.1 Drive shaft torque. The torque required to rotate the drive shaft when all wheel transfer components are engaged shall not exceed 1.0 ounce inches. While exposed at the temperatures specified in 3.3.2, the torque shall not exceed 3.0 ounce inches.

3.3.7.2 Reset shaft torque. The rotational torque, during the reset cycle operation, shall not exceed 30 ounce inches. While exposed at the temperatures specified in 3.2.2, the rotational torque shall not exceed 40 ounce inches.

3.3.7.3 Reset shaft force. When applicable, the direct force applied to the reset shaft to initiate the reset cycle operation shall not exceed 9 lbs. While exposed at the temperature specified in 3.2.2, the applied force shall not exceed 11 lbs.

3.4 Life. The counter shall be capable of rotating in either direction at 1500 RPM continuously and 2500 RPM intermittently for a period of 25 hours and shall reset to "3200" successfully for 1500 trials. After which it shall be exposed to a salt laden atmosphere consisting of 5% sodium chloride atomized spray for 48 hours. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a ph between 6.5 and 7.2 at an ambient temperature of 95° ± 2°F. Subsequent to the salt exposure, the counter shall be capable of operating at the above speeds for 10 hours and reset successfully to "3200" for 600 trials. It shall then be exposed for 240 hours at temperatures varying between 82° and 160°F and a relative humidity between 85 and 95%. After the humidity exposure, the counter shall be capable of operating at the aforementioned speeds for 15 more hours and reset to "3200" successfully for an additional 900 trials. Subsequent to the successful conclusion of the above conditions and exposures, the counter shall meet the requirements of 3.3.7 at standard ambient temperatures.

3.5 Workmanship. The counter shall meet the workmanship requirement of MIL-F-13926. The black background of the mask and numbered wheels shall be free of scratches and blemishes. There shall be no evidence of oil or grease deposits on the periphery of the numbered wheels resulting from counter operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of

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the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection

b. Quality conformance inspection

4.3 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval test shall consist of three (3) completely assembled units. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The sample shall be subjected to the examinations and tests of Tables I and II.

4.3.1 Government testing. When the Government is responsible for conducting first article approval test, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.3.2 Contractor testing. When the contractor is responsible for conducting first article approval test, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.4 Inspection provisions.

4.4.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.4.2 Examination and test.

4.4.2.1 Component and subassemblies. All components and subassemblies to be inspected in accordance with the inspection provisions contained in the applicable Quality Assurance Provisions of MIL-F-13926.

4.4.2.2 Quality conformance inspection. Subsequent to first article approval, examination and test related to Section 3 shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II, except that the speed test of 4.6.3.3 shall be performed on a 100% basis. Examination tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I. Classification of defects

Class Critical: None Major: AQL 1.0% Defective		Requirement	Test procedure
101	General description	3.1.1	4.6.1.2
102	Fabrication	3.1.2	4.6.1.1
103	Finish-digits, index, graduations	3.1.7.1 & 3.1.7.3	4.6.1.5
104	Finish of exterior	3.1.7.2 & 3.1.7.3	4.6.1.5
105	Digit & graduation data	3.1.3	4.6.1.3

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TABLE I. Classification of defects - Continued

106	Digit alignment	3.3.3	4.6.3.2
107	Vibration	3.2.3	4.6.2.2
108	Operation	3.3.1	4.6.3.1
109	Drive shaft operation	3.3.2	4.6.3.1
110	Reset shaft operation	3.3.5	4.6.3.4
112	Torque(static @ 60° - 90°F)	3.3.7.1	4.6.4.1
113	Torque reset shaft (60° - 90°F)	3.3.7.2	4.6.4.2
114	Force reset shaft (60° - 90°F)	3.3.7.3	4.6.4.3
115	Workmanship	3.5	4.6.5
116	Markings (identification)	3.1.9	4.6.1.6
117	Markings (digits & graduations)	3.3.6	4.6.1.3
Minor:	AQL 2.5% Defective		
201	Counter inclosure	3.1.2	4.6.1.1
202	Mask	3.1.4	4.6.1.4

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4.4 Certification. Certification shall include test data and results for all characteristics specified below. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will require a new certification from the contractor.

Characteristics	Test data to comply with
Material	3.1.6.1 & 3.1.6.2
Lubrication	3.1.8
Fungus resistence	3.1.6.3

4.4.5 Special sampling. Sampling inspection for characteristic listed in the following table shall be in accordance with inspection level S3, Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of MIL-STD-105. Samples shall have met all the requirements and tests outlined in Table I prior to being subjected to the following examination and tests.

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Table II. Special sampling

	Characteristic	Requirement	Test procedure
301	Torque drive shaft (-65° +150°F)	3.3.7.1	4.6.4.1
302	Torque reset shaft (-65° + 150°F)	3.3.7.2	4.6.4.2
303	Force reset shaft (-65° + 150°F)	3.3.7.3	4.6.4.3
304	Temperature Test	3.2.1 & 3.2.2	4.6.2.1
305	Shock	3.2.4	4.6.2.3

4.5 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.6 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.6.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.6.2 Contractor furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications forming a part of the contract shall be supplied by the contractor.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.7 Methods of inspection.

4.7.1 Fabrication and general description.

4.7.1.1 Fabrication. All dimensional requirements specified on Drawing D8587543 shall be inspected by means of standard measuring equipment. All measuring equipment used for measuring or gaging shall assure measurements within 90% of tolerance limits. The rigidity of the counter supporting mechanism shall be observed during and after vibration test 4.6.6.2. Use visual inspection techniques to assure sufficient openings have been provided in accordance with 3.1.2. Also use visual inspection tchniques to determine inclosure of counter for compliance with 3.1.3.

4.7.1.2 General description. The counter shall be inspected by visual and tactile inspection techniques in conjunction with the test procedures outlined herein to assure conformance with 3.1.1.

4.7.1.3 Digits and graduations. Use standard measuring equipment and visual inspection techniques to assure digit size and space of graduations and minimum visible length of shortest graduation and index line conform with 3.1.3 when viewed as specified in 3.3.6.

4.7.1.4 Mask. Use visual inspection techniques to determine that individual apertures, mask position and the index position is as specified in 3.1.4.

4.7.1.5 Finishes. All applied finishes, that is, painted, amalgamated in the metal, impregnated, hot stamped or applied chemically shall be in accordance with MIL-STD-171 and visually examined for the appropriate colors specified in 3.1.7.3 to determine conformance with 3.1.7.1 and 3.1.7.2.

4.7.1.6 Markings. Visually inspect part number and serial number for conformance to Military Standard 130 as specified in 3.1.9.

4.7.2 Environmental testing.

4.7.2.1 Temperature test. The units under test shall be subjected to Test Procedure I of MIL-F-13926, however, the temperatures specified in 3.2.1 and 3.2.2 shall apply. When the counters have been exposed and thermally stabilized at the storage temperatures outlined in 3.2.1, the units shall be examined for physical failure. When the counter have been

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exposed and thermally stabilized at the operating temperatures of 3.2.2, the units shall be examined for physical failure and shall be inspected for conformance with 3.3.3, 3.3.5 and 3.3.7 as outlined herein. After having been returned and stabilized at the standard ambient temperatures(73° ± 18°F), the item shall be inspected for conformance with 3.3 and 3.5.

4.7.2.2 Vibration. The units under test shall be subjected to Test Procedure V of MIL-F-13926 except the amplitude frequency, direction and duration specified in 3.2.3 shall apply. At the completion of this test, the units shall be subjected to a visual and tactile inspection to assure conformance with 3.2.3.

4.7.2.3 Shock. The units under test shall be subjected to Test Method 516 of MIL-STD-810. The "G" value and the impacts required shall be as specified in 3.2.4. At the completion of this test, the units shall be subjected to visual and tactile inspection techniques to assure conformance with 3.2.4.

4.7.3 Performance.

4.7.3.1 Operation. A clockwise rotation of the drive shaft shall increase the dial setting. Ten counts of the unit wheel shall increase the second wheel one count. Ten counts on the second wheel shall increase the third wheel on count. Ten counts on the third wheel shall increase the fourth wheel one count. Ten counts on the fourth wheel shall return the counter to the original count as specified in 3.3.1. Counterclockwise rotation of the shaft shall decrease the count in rotation. There shall be no binding or slopping during rotation of the drive shaft as specified in 3.3.2.

4.7.3.2 Digit alignment. Set the unit wheel to any selected setting with the units digit centered symmetrically on the reading line. By means of standard measuring equipment, determine alignment with the other three digits. They shall be aligned within the limits of 3.3.3 afer backlash has been removed.

4.7.3.3 Speed. The counter shall be operated at 500 RPM from a dead stop position for a two (2) minute period in the clockwise direction and for a two (2) minute period in the counterclockwise direction during which time there shall be no evidence of binding, slipping or miscount. The testing medium used to determine proper counter operation shall provide a positive means of detecting momentary high torque caused by mechanical interference in the counter and shall also provide a means of transporting counter input shafts rotation to mil values displayed on the counter under test. The test shall be performed as follows:

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a. Secure the 3200 mil counter to a testing medium.

b. Set the recording test medium to a predetermined setting.

c. Set the counter under test to 3200 mils.

d. Record both readings.

e. Rotate the counter under test at 2500 RPM from a dead stop position in the clockwise or counterclockwise direction for a two (2) minute period. Record both counter readings and determine accuracy of counter under test and note any high torque readings during this period. To simplify calculation, rotate the recording medium to indicate a non-fractional reading. The counter under test must record a reading identical to the calculated reading within the width of one graduate space (.25mil).

f. Rotate the counter under test at 100 RPMs in the same direction through one cycle (999 rotations of the unit wheel) and observe the means for detecting high torque. When abnormal fluctuations of the recording medium are observed, record the counter readings.

g. Remove the counter from the test fixture and test in accordance with 4.6.4.1, except the counter shall be positioned approximately 5 counts prior to any readings recorded in (f) above. Any measurement of torque shall not exceed the limit specified.

h. Repeat steps (a) to (g) inclusive, however, the counter under test shall be rotated in the opposite direction.

4.7.3.4 Reset mechanism. Using tactile and visual inspection techniques determine that the reset mechanism provides the type of activation and also a positive locking action at completion of reset cycle as specified in 3.1.5.

4.7.3.5 Reset operation. The counter shall be set to selected reading within the dial setting range listed below and shall be reset to 3200 with one counterclockwise rotation of the reset shaft as specified in 3.1.5 after each selected setting to determine conformance with the requirements of 3.3.5.

Dial setting range	Number of reset
0000 to 0009	1
0010 to 0090	2
0100 to 0999	10
1000 to 9999	37

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4.7.4 Operability.

4.7.4.1 Torque (drive shaft). Prior to performing static torque of drive shaft, set the units under test to maximum count (9999). Torque test to be performed using standard shop type torque measuring equipment providing readings in ounce - inches to an accuracy specified in 4.5. Maximum torque readings shall not exceed the limits outlined in 3.3.7.1 when the unit test has been exposed and stabilized at the temperature outlined in 3.3.7.1.

4.7.4.2 Torque (reset shaft). Rotational torque of the reset shaft in the unlocked position shall be measured using standard shop type torque measuring equipment providing readings in ounce - inches and having an accuracy within the limits outlined in 4.5. Maximum torque readings shall not exceed the limits outlined in 3.3.7.2 when the unit under test has been exposed and stabilized at the temperature outlined in 3.3.7.2.

4.7.4.3 Force (reset shaft). Force required to unlock the reset shaft when applicable shall be measured using standard shop type push-pull measuring equipment providing readings in pounds and having an accuracy within the limits outlined in 4.5. Maximum push-pull readings shall not exceed the limits outlined in 3.3.7.3 when the unit under test has been exposed and stabilized at the temperatures outlined in 3.3.7.3.

4.7.5 Workmanship. Using visual and tactile inspection techniques determine that the quality of workmanship complies with the requirement of 3.5.

4.7.6 Life test. In addition to the first article test, the life test shall be performed on three (3) units randomly selected at the 25%, 50% and 75% point of production except that all the selected production samples shall be subjected to only 10% of the time duration for the life test. They shall have complied with all characteristics specified in Tables I and II, however, 4.6.6.3 and 4.6.6.6 shall not apply to the production test samples. The test equipment used to perform the life test shall conform to the accuracies specified in 4.5.

The test shall be performed as follows:

a. Rotate the counter drive shaft 1500 RPM for 358 seconds and at 2500 RPM for 2 seconds. Cycle time 6 minutes. Repeat the cycle 125 times in the clockwise and then 125 times in the counterclockwise.

b. After each 5 hours of rotating the drive shaft, the counter shall be given 300 reset trials. Total 1500 resets.

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c. Subsequent to completely satisfying the 25 hours of rotation and 1500 reset trials all samples shall be subjected to Salt Fog Test specified in Method 509 of MIL-STD-810, however, the values specified in 3.4 shall apply.

d. Upon successfully completing all previous tests, the counter shall be given additional drive shaft rotations at the speeds specified in 3.4. Cycle time to be 6 minutes (see 4.6.6.1). Repeat the cycle 50 times in the clockwise and then 50 times in the counterclockwise direction. Total cycling time 10 hours.

e. After each hour of the previous test for rotation, the counter shall be subjected to 60 reset trials. Total 600 reset trials.

f. Subsequent to satisfactorily completing all previous tests, the counter shall be subjected to the humidity test specified in Test Procedure I of Test Method 507 of MIL-STD-810.

g. Resubmit the counter to an additional operational test at the speeds specified in 3.4. Cycle time to be 6 minutes (see 4.6.6.1). Repeat the cycle 75 times in the clockwise and then 75 times in the counterclockwise direction. Total cycling time 15 hours.

h. Upon completing each 50 cycles specified in 4.6.6.7, the counter shall satisfactorily reset 300 times. Total 900 resets.

i. The total number of reset failures permitted, that is, requiring more than one (1) turn of the reset shaft shall not exceed the limits specified in 3.1.5.

j. When the counter has successfully complied with the requirements of 3.4 when inspected either as outlined in 4.6.6 or using an alternate method equal to or exceeding the severity of inspection described herein, the counter shall be reinspected for torque reading at ambient temperature, as outlined in 4.6.4.

5. PACKAGING

5.1. Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 8587543. The level of protection shall be as specified in the procurement document (see 6.2).

6. NOTES

6.1 Intended use. The counter covered by this specification is intended for use in panoramic telescopes for recording azimuth readings.

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6.2 Acquisition requirements. Procurement documents should specify:

a. Title, date and number of this specification.

b. Serial numbers to be used for identification marking

c. Applicable packaging data sheet number.

d. Selection of applicable levels of preservation and packaging.

e. Requirements for first article testing.

f. Contract data requirements for submission of inspection equipment designs conforming to Data Item Description DI-R-10054.

6.3 Inspection equipment design.

6.3.1 Submission of designs for approval. Contractor designs for final acceptance inspection shall be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, U.S. Army Armament, Research, Development and Engineering Center, ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply. When the contractor submits inspection equipment designs to the government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, model number, etc.).

c. The designs remaining to be submitted and the expected date of submittal.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.5 Subject term (key word) listing.

Counter
Digits
Index
Graduations
Fire Control

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament, Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1290-A432)

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